

Yuxiang Wan

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🌐 Personal Page

🌐 LinkedIn

🐙 GitHub

Education

University of Illinois at Urbana-Champaign, B.S. in Computer Engineering	Sep. 2023 – May 2027
GPA: 4.00/4.00	(expected)
Zhejiang University, B.E. in Electrical and Computer Engineering	Sep. 2023 – May 2027
GPA: 3.96/4.00, Rank: 1/70	(expected)
Selective coursework:	
◦ Data Structures (A+)	◦ Database Systems (A)
◦ Digital Signal Processing (A+)	◦ Probability (A+)
	◦ Computer Systems (A+)
	◦ Deep Generative Models (A+)

Research Interest

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- Building capable and efficient **AI agents** grounded in language and vision, with a current focus on **computer-use and GUI agents**, **LLM inference and decoding**, and structured in-context learning.
 - More broadly, I am interested in methods that improve the efficiency, generalization, and decision-making ability of large models when deployed in interactive, real-world environments.

Research Experience

Graph-Structured In-Context Learning	AI-Net Lab, Zhejiang University	Jun. 2024 – Apr. 2025
Advisor: Prof. Yang Yang		
◦ Aimed to improve in-context learning by introducing structural relationships among demonstration candidates and the test query, addressing the limitation that existing methods select examples independently.		
◦ Proposed CONTEXTG: builds a learnable graph over retrieved examples and the query, uses PageRank-based filtering to select structurally important demonstrations, and encodes them at node/path/subgraph levels to capture multi-hop dependencies.		
◦ Outperformed SOTA by 2.74% (avg.); best average performance in cross-task transfer.		
Efficient Inference for Diffusion-based Language Models	UIUC	Feb. 2026 – Present
Advisor: Prof. Fan Lai		
◦ Investigating efficient inference for discrete diffusion LLMs (dLLMs), where inter-step decoding (ISD) pipelines denoising but suffers from a <i>horizon problem</i> due to its flat propose layout.		
◦ Proposed: (1) Tree-Structured ISD —entropy-guided branching to improve tokens-per-forward-pass; (2) Content-Adaptive Stride —dynamically adjusting propose window width based on local token difficulty.		
◦ Literature survey and proposal completed; implementation in progress.		

Projects

ECE 391: UNIX-like Operating System Kernel	C, RISC-V Assembly	Fall 2025
◦ Built a UNIX-like OS kernel on RISC-V from scratch: implemented Sv39 virtual memory (3-level page tables, demand paging), process abstraction (fork, exec), and preemptive multitasking with round-robin scheduling		
◦ Developed VIRTIO block device driver and KTFS filesystem with 64-block write-back cache, supporting CRUD operations, indirect/doubly-indirect block addressing, and persistent storage		
◦ Implemented 15+ system calls (fork, exec, read, write, pipe, etc.) and a functional shell with I/O redirection and piping; wrote Unix utilities (cat, ls, wc, xargs) for command-line interaction		

Selected Honors and Awards

National Scholarship (Top 1%)	2024, 2025
Dean's List, University of Illinois at Urbana-Champaign	2024, 2025
First Prize Scholarship for Academic Excellence, ZJUI	2025

Teaching Experience

Teaching Assistant, Computer Systems & Programming

Spring 2025

- Led office hours and discussion sessions for 300+ students; provided guidance on C programming and debugging
- Refactored and automated autograder scripts, reducing grading time and improving feedback efficiency

Teaching Assistant, Linear Algebra

Fall 2024

- Conducted weekly problem-solving sessions for 100+ students; organized comprehensive final review session